
Algorithm 7: Smart-Greedy Rollout Probe with Tabu List

Input:

- L_{start} . The starting label $(v, t, C, \text{pred}, \text{start_time})$
- E_s . Set of scheduled edges $(v, u, \text{dep}, \text{arr})$
- E_t . Set of transfer edges (v, w, Δ)
- $\text{country}(v)$. Country associated with node v

Output:

- L_{final} . The final label reached when the 24-hour clock expires

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1:  $L \leftarrow L_{\text{start}}$ 
2:  $\text{RolloutVisitedNodes} \leftarrow \{L.\text{node}\}$  // Tabu list to prevent all local loops
3: // Repeat until no more valid moves within 24 hours
4: repeat
5:    $\text{BestNodeArrivals} \leftarrow \text{Empty Map}$  // Maps a node 'u' to the earliest arrival time 'tnew' and
   departure 'dep'
6:   // 1. Collect ONLY the earliest arrival for each neighboring node
7:   for all  $(L.\text{node}, u, \text{dep}, \text{arr}) \in E_s$  do
8:     if  $\text{dep} \geq L.t$  &  $\text{arr} - L.\text{start\_time} \leq 1440$  then
9:       if  $u \notin \text{BestNodeArrivals}$  or  $\text{arr} < \text{BestNodeArrivals}[u].\text{arr}$  then
10:         $\text{BestNodeArrivals}[u] \leftarrow (\text{dep}, \text{arr}, \text{country}(u))$ 
11:   for all  $(L.\text{node}, w, \Delta) \in E_t$  do
12:      $t_{\text{new}} \leftarrow L.t + \Delta$ 
13:     if  $t_{\text{new}} - L.\text{start\_time} \leq 1440$  then
14:       if  $w \notin \text{BestNodeArrivals}$  or  $t_{\text{new}} < \text{BestNodeArrivals}[w].\text{arr}$  then
15:         // For transfers, dep and arr are both tnew
16:          $\text{BestNodeArrivals}[w] \leftarrow (t_{\text{new}}, t_{\text{new}}, \text{country}(w))$ 
17:   if  $\text{BestNodeArrivals}$  is empty then
18:     break // No valid forward moves left
19:   // 2. Assign heavily biased weights to the deduplicated list
20:    $\text{ValidSteps} \leftarrow \emptyset$ 
21:    $\text{curr\_country} \leftarrow \text{country}(L.\text{node})$ 
22:   for all  $u \in \text{BestNodeArrivals}$  do
23:      $(\text{dep}, \text{arr}, c_{\text{new}}) \leftarrow \text{BestNodeArrivals}[u]$ 
24:      $\Delta t \leftarrow \max(1, \text{arr} - L.t)$  // Total time consumed by this step
25:     if  $c_{\text{new}} \notin L.C$  then
26:       // UNVISITED COUNTRY: Absolute highest priority.
27:        $\text{base\_weight} \leftarrow 100,000$ 
28:     else if  $c_{\text{new}} \neq \text{curr\_country}$  then
29:       // TRANSIT VIA OTHER COUNTRY: Crucial for escaping geographical traps.
30:        $\text{base\_weight} \leftarrow 1,000$ 
31:     else
32:       // DOMESTIC TRANSIT: Just used to get to hubs.
33:        $\text{base\_weight} \leftarrow 1$ 
34:      $\text{weight} \leftarrow \text{base\_weight} / \Delta t$  // Linear time penalty
35:     // THE TABU PENALTY: Prevent wandering in local loops
36:     if  $u \in \text{RolloutVisitedNodes}$  then
37:        $\text{weight} \leftarrow \text{weight} \times 0.0001$ 
38:      $\text{ValidSteps} \leftarrow \text{ValidSteps} \cup \{(u, \text{arr}, c_{\text{new}}, \text{weight})\}$ 

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39: | // 3. Roll the weighted dice
40: |  $s_{\text{chosen}} \leftarrow \text{WeightedRandomChoice}(\text{ValidSteps})$ 
41: | // 4. Update State
42: |  $C' \leftarrow L.C \cup \{s_{\text{chosen}}.c_{\text{new}}\}$ 
43: |  $\text{RolloutVisitedNodes} \leftarrow \text{RolloutVisitedNodes} \cup \{s_{\text{chosen}}.u\}$ 
44: |  $L \leftarrow (s_{\text{chosen}}.u, s_{\text{chosen}}.\text{arr}, C', \text{pred} = L, \text{start\_time} = L.\text{start\_time})$ 
45: until false
46: return  $L$ 

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